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June 22, 2026

The Editor-in-Chief
Remote Sensing
MDPI

Dear Editor,

We are pleased to submit our manuscript, *"Harmonized Sentinel-2/Landsat Retrieval of Water Transparency in Lake Titicaca Using Machine Learning: A Transferable Validation and Retrievability Protocol for High-Altitude Lakes (2013–2024)"*, for consideration as a research article in *Remote Sensing*.

The study sits squarely within the journal's scope—machine-learning retrieval of water-quality variables from harmonized optical satellite missions. We present, to our knowledge, the first multi-mission harmonized retrieval of water transparency (Secchi depth) for a high-altitude Andean lake (3,810 m a.s.l.), built entirely on **real in-situ measurements** (881 field observations, 1,002 satellite–field match-ups, 2013–2024) sampled at their true coordinates across the whole lake. It directly complements recent work on Lake Titicaca published in *Remote Sensing* using global products (Maligaya et al., 2024), to which we add harmonized multi-mission retrieval, local in-situ validation, and an explicit retrievability assessment.

We believe this work makes four distinctive contributions. (1) **A transferable validation hierarchy:** beyond station-wise cross-validation and a temporal hold-out, we introduce a leave-one-zone-out test showing that, although random and station-wise validation are identical (no spatial-autocorrelation leakage), extrapolation to an unseen optical regime collapses R^2 toward zero—a loss of transferability that random splits conceal. (2) **Calibrated uncertainty:** we attach per-pixel split-conformal prediction intervals (empirical coverage 89.8% at a 90% nominal level). (3) **Physical interpretability:** SHAP shows the retrieval is driven by green/blue reflectance ratios, consistent with water-clarity optics. (4) **Transparent delimitation of limits:** we report explicitly that chlorophyll-a, suspended solids and thermal temperature are *not* retrievable in this oligotrophic regime—an honest operational envelope that cautions against transferring turbid-water algorithms to clear high-altitude lakes. A lake-wide transparency map demonstrates operational utility.

All data and code are openly available, and the full pipeline is reproducible from public satellite archives and published institutional monitoring records. The manuscript is original, not under consideration elsewhere, and all authors approve the submission. We have no conflicts of interest to declare.

Thank you for considering our work.

Sincerely,

Richar Andre Vilca-Solorzano
(on behalf of all authors)